

Complexity of Cycle Transverse Matching Problems

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Abstract. The stable transversal problem for a fixed graph H asks whether a graph contains a stable set that meets every induced copy of H in the graph. Stable transversal problems generalize several vertex partition problems and have been studied for various classes of graphs. Following a result of Farrugia, the stable transversal problem for each C_ℓ with $\ell \geq 3$ is NP-complete. In this paper, we study an ‘edge version’ of these problems. Specifically, we investigate the problem of determining whether a graph contains a matching that meets every copy of H . We show that the problem for C_3 is polynomial and for each C_ℓ with $\ell \geq 4$ is NP-complete. Our results imply that the stable transversal problem for each C_ℓ with $\ell \geq 4$ remains NP-complete when it is restricted to line graphs. We show by contrast that the stable transversal problem for C_3 , when restricted to line graphs, is polynomial.

Keywords: Stable transversal problem, transverse matching problem, algorithm, complexity.

1 Introduction

For a fixed graph H , an H -transversal of a graph G is a vertex set T that *meets* every induced copy of H in G ; in other words, $G - T$ does not contain H as an induced subgraph. When such a set T is a stable set, it is called a *stable H -transversal* of G . While every graph has an H -transversal, stable H -transversals may not exist in a graph. The *stable H -transversal problem* $\text{ST}(H)$ for a fixed H asks whether a graph has a stable H -transversal.

It is easy to see that a graph has a stable K_2 -transversal if and only if it is bipartite and it has a stable $\overline{K_2}$ -transversal if and only if it is a split graph (cf. [4]). As both bipartite and split graphs are recognizable in polynomial time, $\text{ST}(H)$ is polynomial when H has at most two vertices. Farrugia [3] proved that $\text{ST}(H)$ is NP-complete when H has at least three vertices and is connected. It

follows that $\text{ST}(C_\ell)$ is NP-complete when $\ell \geq 3$, where C_ℓ denotes the cycle with ℓ vertices. For graphs which have at least three vertices and are disconnected, there are polynomial as well as NP-complete cases, but the dichotomy of stable transversal problems has not been determined. Stable transversal problems have also been studied for various classes of graphs, cf. [7,9].

In this paper, we investigate an ‘edge version’ of stable transversal problems. The *H-transverse matching problem* $\text{TM}(H)$ for a fixed graph H asks whether a graph G has a matching M which meets every (not necessarily induced) copy of H in G , i.e., H is not a subgraph of $G - M$. Such a matching M will be called an *H-transverse matching*. We shall prove the following:

Theorem 1. *$\text{TM}(C_\ell)$ is polynomial when $\ell = 3$ and NP-complete otherwise.*

Given a graph $G = (V, E)$, the *line graph* $L(G)$ of G is the graph with vertex set E and two vertices adjacent in $L(G)$ if and only if the two corresponding edges have a common endvertex in G . With the sole exception of $H = C_3$, the transverse matching problem $\text{TM}(H)$ is equivalent to the stable transversal problem $\text{ST}(L(H))$ for line graphs. Since $L(H) = C_\ell$ if and only if $H = C_\ell$ for each $\ell \geq 4$, Theorem 1 implies that $\text{ST}(C_\ell)$ remains NP-complete for line graphs for each $\ell \geq 4$.

Due to the fact that C_3 is the line graph of two different graphs C_3 and $K_{1,3}$, $\text{ST}(C_3)$ is not equivalent to $\text{TM}(C_3)$ for line graphs. However, we show that determining whether a graph G has a matching M which meets every copy of C_3 and $K_{1,3}$ in G is polynomial. Hence the problem $\text{ST}(C_3)$ is polynomial for line graphs and therefore we also obtain the following:

Theorem 2. *When restricted to line graphs, $\text{ST}(C_\ell)$ is polynomial for $\ell = 3$ and NP-complete otherwise.*

We remark that the polynomial case stated in Theorem 2 is in sharp contrast to the fact that $\text{ST}(C_3)$ is NP-complete in general.

2 $\text{TM}(C_3)$ and $\text{ST}(C_3)$ for Line Graphs are Polynomial

A *paw* is the graph consisting of a triangle and an edge sharing a common vertex. Call a matching M *pawssible* if, for every paw consisting of triangle xyz and edge xw , $xw \in M$ implies $yz \in M$. The lemma below follows immediately from the fact that any C_3 -transverse matching contains at least one of the three edges in each triangle and is pawssible.

Lemma 3. *If a graph G has a C_3 -transverse matching, then at least one of the three edges in each triangle is contained in a pawssible matching. $\square$*

In polynomial time we can check whether an edge e of G is contained in a pawssible matching and find one if it exists. This can be done by letting $F_e = \{e\}$ initially, then adding edges one by one to F_e which are ‘forced by’ paws, and finally checking if F_e is a matching. In the case when F_e is a matching, it is also a minimal pawssible matching containing e .

Lemma 4. *Suppose that M is a pawssible matching and e is an edge in G . If both $M \cup \{e\}$ and F_e are matchings, then $M \cup F_e$ is a pawssible matching where F_e is defined as above.*

Proof. From the definition of F_e , $M \cup F_e$ may be viewed as the set obtained from M by including e initially and then adding edges which are ‘forced by’ paws. The process of adding edges ensures that $M \cup F_e$ is pawssible. We claim that it produces a matching. Indeed, suppose that a paw consists of triangle xyz and edge xw such that $M \cup \{xw\}$ is a matching. Since M is pawssible, M does not contain any edge incident with y or z as otherwise M would contain xz or xy , contradicting the assumption that $M \cup \{xw\}$ is a matching. Hence $M \cup \{xw, yz\}$ is also a matching. This means that the process of adding edges always produces matchings. So $M \cup F_e$ is a matching. $\square$

Theorem 5. *A graph G has a C_3 -transverse matching if and only if every triangle has an edge that is contained in a pawssible matching.*

Proof. The necessity is treated in Lemma 3. For sufficiency, suppose that every triangle has an edge that is contained in a pawssible matching. The assumption implies in particular that every triangle has an edge e for which F_e is a pawssible matching.

We apply the following algorithm to construct a set M of edges in G : Initially, $M = \emptyset$. As long as $G - M$ contains a triangle, find an edge e which is contained in a triangle in $G - M$ and for which F_e is a matching and enlarge M to include F_e .

Clearly, the set M contains at least one edge from each triangle in G . Lemma 4 ensures that M obtained by the algorithm is a matching. Therefore M is a C_3 -transverse matching in G . $\square$

Theorem 5 and its proof suggest a polynomial time algorithm which determines if a graph G has a C_3 -transverse matching and constructs one if it exists: We first compute F_e for each edge e of G as described above. If for some triangle xyz , none of F_{xy}, F_{yz}, F_{xz} is a matching, then G does not have a C_3 -transverse matching according to Theorem 5. Otherwise, the algorithm in the proof of Theorem 5, which can be implemented in polynomial time, constructs a C_3 -transverse matching.

Corollary 6. *$TM(C_3)$ is polynomial.* $\square$

Next we show how to determine, in polynomial time, whether a graph has a matching which meets every copy of C_3 and of $K_{1,3}$. It is well-known that a perfect matching (if exists) can be found in polynomial time (cf. [2,8]). We can apply such an algorithm to determine if a graph has a matching covering specified vertices.

Lemma 7. *Given a graph G and vertex set S , it can be determined in polynomial time whether G has a matching covering every vertex of S .*

Proof. We show how to convert the problem in the lemma into the problem of finding a perfect matching. Let G' be a (disjoint) copy of G with $f : G \rightarrow G'$ an isomorphism. Let G^* be obtained from $G \cup G'$ by adding the edge $vf(v)$ if $v \in V(G) - S$. If M is a matching of G covering every vertex of S , then $M \cup f(M) \cup \{vf(v) : v \text{ is not covered by } M\}$ is a perfect matching of G^* . Conversely, any perfect matching of G^* restricts to a desired matching of G . Thus G has a matching covering every vertex of S if and only if G^* has a perfect matching. $\square$

Clearly, no graph of maximum degree ≥ 4 contains a matching which meets every copy of $K_{1,3}$. So we may restrict the search of a matching among graphs of maximum degree at most three (i.e., *subcubic* graphs).

Proposition 8. *There exists a polynomial time algorithm to determine whether a subcubic graph has a matching that meets every copy of C_3 and of $K_{1,3}$.*

Proof. We may assume without loss of generality that G is connected. We show how to reduce the problem in the proposition to the problem of determining whether a graph has a matching covering specified vertices, which can be solved in polynomial time according to Lemma 7. Specifically, we will construct a subcubic graph G' and a set $S \subseteq V(G')$ such that G has a desired matching if and only if G' has a matching covering S .

Initially, we let $G' = G$ and let S consist of all vertices of degree 3 in G' . If G' has no triangle, then G' together with S satisfy the desired properties. So assume that G' contains triangles. We modify G' and the set S recursively for every triangle. Each modification on a triangle results in a subcubic graph and maintains the property that S consists of all vertices of degree three. Moreover, any matching covering S in the resulting graph must meet the triangle.

Let abc be a triangle in G' and a', b', c' be the only other neighbours of a, b, c respectively, if they exist. If a', b', c' exist and are the same vertex, then the graph is K_4 which has a desired matching. If, without loss of generality, $a' = b'$ exist and are distinct from c' (if it exists), then we do nothing. Any matching covering a, b (which are in S) in the resulting graph meets the triangle abc . On the other hand, if a', b', c' exist and are distinct we contract the triangle, identifying the vertices a, b, c in G' and in S . Finally, if a', b', c' are distinct but do not all exist, we contract the triangle but also remove the identified vertex from S . Clearly, G has a desired matching if and only if the resulting graph has a desired matching.

By repeating the above reduction for every triangle in G' , we either obtain K_4 (in which case G has a desired matching) or a graph G' and a set S such that G has a desired matching if and only if G' has a matching covering S . $\square$

Corollary 9. *$ST(C_3)$ for line graphs is polynomial.* $\square$

3 $TM(C_4)$ is NP-Complete

In this section, we show that $TM(C_4)$ is NP-complete even when it is restricted to bipartite graphs.

Proposition 10. *TM(C_4) for bipartite graphs is NP-complete.*

Proof. We reduce from 4-SAT, which is well-known to be NP-complete [5]. Let $\mathcal{C} = \{c_1, \dots, c_q\}$ be a set of clauses each consisting of four variables from $\mathcal{A} = \{a_1, a_2, \dots, a_p, \overline{a_1}, \dots, \overline{a_p}\}$. A valid truth assignment for $\mathcal{C}$ is a function $f : \mathcal{A} \rightarrow \{0, 1\}$ such that $f(a_i) \neq f(\overline{a_i})$ and $f^{-1}(1) \cap c_j \neq \emptyset$ for every $i = 1, \dots, p$ and $j = 1, \dots, q$. In other words, $\overline{a_i}$ is the negation of a_i and every clause has at least one true variable. We construct in polynomial time a bipartite graph H which admits a C_4 -transverse matching if and only if such a satisfiability function exists.

There are two main gadgets in our reduction: one corresponding to each variable and one corresponding to each clause. The variable gadgets (see Fig. 1, upper left) are constructed as follows. Associate each a_i , $1 \leq i \leq p$ with a path $w_0^i w_1^i w_2^i w_3^i w_4^i$; three vertices v_1^i, v_2^i, v_3^i adjacent to w_0^i and w_2^i , w_1^i and w_3^i , and w_2^i and w_4^i , respectively; three more vertices u_1^i, u_2^i, u_3^i ; and nine vertices $v_{k,1}^i, v_{k,2}^i, v_{k,3}^i$ each adjacent to the respective u_k^i and v_k^i . It is easy to check that the gadget admits C_4 -transverse matchings containing $w_0^i w_1^i$ and $w_2^i w_3^i$, or containing $w_1^i w_2^i$ and $w_3^i w_4^i$, and that any such matching contains one of those pairs.

The clause gadgets (see Fig. 1), on the other hand, are somewhat simpler than the variable gadgets. Each clause $c_j \in \mathcal{C}$ is associated with a $C_8 : s_1^j t_1^j s_2^j t_2^j s_3^j t_3^j s_4^j t_4^j$ and two additional vertices r_1^j, r_2^j adjacent to t_1^j, t_2^j, t_3^j and t_2^j, t_3^j, t_4^j , respectively. It turns out that this gadget has a C_4 -transverse leaving any three of $s_1^j, s_2^j, s_3^j, s_4^j$ uncovered, but no C_4 -transverse matching leaving all four uncovered.

We complete the construction of H by connecting the clause gadgets to the variable gadgets. Let $c_j = \{a_{j1}, a_{j2}, a_{j3}, a_{j4}\}$. If $a_{jk} = a_i$, add two new vertices x_k^j, y_k^j and edges $x_k^j y_k^j, s_k^j y_k^j, w_2^i y_k^j, w_1^i x_k^j$ and $w_3^i s_k^j$ (thus $w_1^i w_2^i w_3^i s_k^j y_k^j x_k^j$ is a C_6 : see Fig. 1). On the other hand, if $a_{jk} = \overline{a_i}$, do the same thing with the roles of w_1^i and w_3^i reversed.

Suppose the resulting graph H has a C_4 -transverse matching M . We interpret M as a truth assignment f according to its intersection with each variable gadget, setting

$$f(a_i) = \begin{cases} 0 & \text{if } w_1^i w_2^i, w_3^i w_4^i \in M \\ 1 & \text{if } w_0^i w_1^i, w_2^i w_3^i \in M \end{cases}$$

and $f(\overline{a_i}) \neq f(a_i)$. We must show that $f^{-1}(1) \cap c_j \neq \emptyset$ for all $j = 1, \dots, q$.

The above discussion of the clause gadgets implies that, for a fixed j , the matching M contains at most three of the four edges $s_k^j y_k^j$, $k = 1, 2, 3, 4$; as M must cover at least one s_k^j with an edge from the clause gadget associated with c_j . Suppose that $a_i = a_{jk} \in c_j$. The above discussion of the variable gadgets implies that neither $w_2^i y_k^j$ nor $w_3^i s_k^j$ is in M , and since M covers the $C_4 : w_2^i w_3^i s_k^j y_k^j$ it must be the case that $w_2^i w_3^i \in M$. Consequently, $a_i \in f^{-1}(1) \cap c_j$. A similar argument applies when $\overline{a_i} = a_{jk} \in c_j$, showing that $\overline{a_i} \in f^{-1}(1) \cap c_j$. It follows that f is a satisfiability function.

Conversely, suppose that f is a satisfiability function for the clauses $\mathcal{C}$. Then, as mentioned above, we can extend

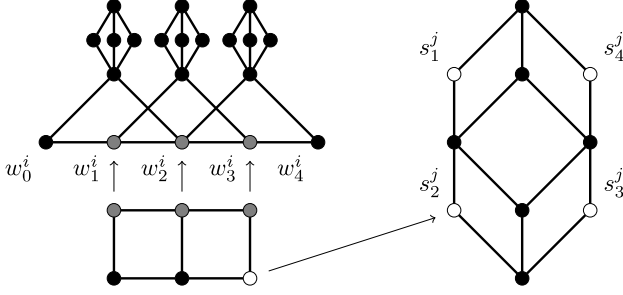


Fig. 1. Connecting the gadget corresponding to clause $c_j = \{a_{j1}, a_{j2}, a_{j3}, a_{j4}\}$ (right) with the gadget corresponding to the variable $a_i = a_{j2}$ (upper left)

$$\begin{aligned}
 M' = & \{w_1^i w_2^i, w_3^i w_4^i : f(a_i) = 0\} \\
 & \cup \{w_0^i w_1^i, w_2^i w_3^i : f(a_i) = 1\} \\
 & \cup \{y_k^j s_k^j : f \text{ maps the } k\text{th variable in clause } c_j \text{ to } 0\}
 \end{aligned}$$

to a C_k -transverse matching of H , using the fact that M' leaves at least one s_k^j uncovered for every j , and the clause gadget has a C_k -transverse matching leaving the other three uncovered.

Finally, H has bipartition $(X, V(H) \setminus X)$, where

$$\begin{aligned}
 X = & \{w_0^i, w_2^i, w_4^i, v_{k,1}^i, v_{k,2}^i, v_{k,3}^i : i = 1, \dots, p; k = 1, 2, 3\} \\
 & \cup \{x_k^j, s_k^j : j = 1, \dots, q; k = 1, 2, 3, 4\}.
 \end{aligned}$$

The construction of H takes $O(p+q)$ time, so this reduction is polynomial. Therefore, the C_4 -transverse matching problem for bipartite graphs is NP-complete. $\square$

4 $\text{TM}(C_\ell)$ is NP-Complete for $\ell \geq 5$

A broadly similar approach can be taken to show that $\text{TM}(C_\ell)$ is NP-complete for each $\ell \geq 5$. As the variable gadgets are slightly different depending on whether ℓ is odd or even, we treat these cases separately. However, both cases depend on the following useful structure.

A *u-v accordion path* consists of edge-disjoint $u-v$ paths $ux_1w_1x_2 \dots w_kx_{k+1}v$, $uy_1w_1y_2 \dots w_{k-1}y_kv$, and $uy_1w_1y_2 \dots w_ky_{k+1}v$, each having the same (even) length and sharing every second vertex. An example is given in Fig. 2. The *length* of an accordion path is the length (i.e. number of edges) of any of the paths comprising it.

The importance of accordion paths comes from the following observation: any matching in a $u-v$ accordion path fails to cover some $u-v$ path of the same length k . More importantly, if a graph contains such an accordion path and an additional $u-v$ path P of length $\ell - k$, then any C_ℓ -transverse matching contains at least one of the edges of P . We will find this property very useful when constructing our gadgets.

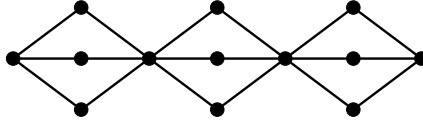


Fig. 2. An accordion path of length 6

Proposition 11. $TM(C_\ell)$ for bipartite graphs is NP-complete when $\ell \geq 6$ is even.

Proof. The reduction is from $\frac{\ell}{2}$ -SAT, using a similar strategy as the one in the previous section. This time, the clause gadget corresponding to each c_j is simply a $C_\ell : s_1^j t_1^j \dots s_{\ell/2}^j t_{\ell/2}^j$. By definition, any C_ℓ -transverse matching of the resulting graph contains at least one edge of this cycle and hence covers at least one s_k^j . The variable gadget corresponding to a_i consists of a path $w_0^i w_1^i w_2^i w_3^i w_4^i$ and three disjoint accordion paths of length $\ell - 2$ from w_0^i to w_2^i , from w_1^i to w_3^i , and from w_2^i to w_4^i . The only minimal C_ℓ -transverse matchings of this gadget are $\{w_0^i w_1^i, w_2^i w_3^i\}$ and $\{w_1^i w_2^i, w_3^i w_4^i\}$.

We connect the variable gadgets to the clause gadgets as follows. If $c_j = \{a_{j1}, a_{j2}, a_{j3}, a_{j4}\}$ and $a_{jk} = a_i$, add three new vertices x_k^j, y_k^j, z_k^j , edges $x_k^j y_k^j, s_k^j y_k^j, w_0^i x_k^j, w_1^i z_k^j, w_3^i s_k^j$, and an accordion path of length $\ell - 4$ from z_k^j to y_k^j . Likewise, if $a_{jk} = \bar{a}_i$, do the same except adding edges $w_1^i s_k^j$ and $w_3^i x_k^j$ instead of $w_0^i s_k^j$ and $w_1^i x_k^j$. This process is illustrated in Fig. 3.

As in the previous section, we can interpret a C_4 -transverse matching M of the resulting bipartite graph H as a truth assignment f by

$$f(a_i) = \begin{cases} 0 & w_1^i w_2^i, w_3^i w_4^i \in M \\ 1 & w_0^i w_1^i, w_2^i w_3^i \in M \end{cases}$$

and $f(\bar{a}_i) \neq f(a_i)$. A similar argument to that in the proof of Proposition 10 shows that f is a satisfiability function for $\mathcal{C}$. Likewise, a satisfiability function can be used to construct a C_ℓ -transverse matching for H . The graph H is easily seen to be bipartite. $\square$

When ℓ is odd, we can use a nearly identical reduction and proof to show that $TM(C_\ell)$ is NP-complete. The graph produced by the reduction is not bipartite in this case, for any matching in a bipartite graph is C_ℓ transverse for odd ℓ .

Proposition 12. $TM(C_\ell)$ is NP-complete when $\ell \geq 5$ is odd.

Proof. We only describe the construction of H from an instance $\mathcal{C}$ of $\frac{(k+1)}{2}$ -SAT. As in the proof of Proposition 11, the gadget corresponding to each clause c_j is a cycle $C_\ell : s_1^j t_1^j s_2^j t_2^j \dots s_{(k+1)/2}^j t_{(k+1)/2}^j$. Each variable a_i corresponds to a path $w_0^i w_1^i w_2^i w_3^i w_4^i$ in such a way that any C_ℓ -transverse matching contains either $\{w_0^i w_1^i, w_2^i w_3^i\}$ or $\{w_1^i w_2^i, w_3^i w_4^i\}$; to ensure this property we add vertices u^i, v^i ,

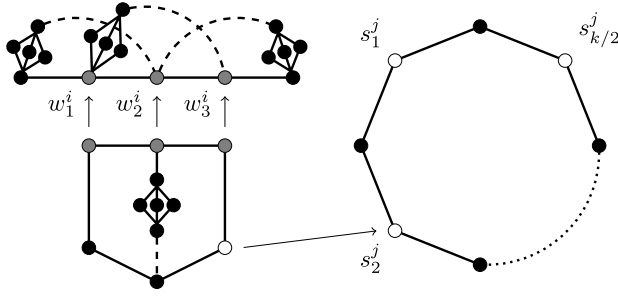


Fig. 3. Connecting the cycle corresponding to clause c_j to the variable gadget corresponding to a_i . Dashed lines indicate accordion paths.

edges $u^i w_2^i, u^i w_3^i, u^i w_4^i$ and $u^i v^i$, an accordion path of length $\ell - 1$ from u^i to v^i , and accordion paths of length $\ell - 3$ from w_0^i, w_1^i , and w_2^i to u^i . (See Fig. 4). As desired, the minimal C_ℓ -transverse matchings of this gadget are $\{w_0^i w_1^i, w_2^i w_3^i, u^i v^i\}$ and $\{w_1^i w_2^i, w_3^i w_4^i, u^i v^i\}$.

The clause gadgets are connected with the appropriate variable gadgets in nearly the same way as in Proposition 11. If $c_j = \{a_{j1}, a_{j2}, a_{j3}, a_{j4}\}$ and $a_{jk} = a_i$, add two new vertices x_k^j, y_k^j , edges $x_k^j y_k^j, s_k^j y_k^j, w_0^i x_k^j, w_3^i s_k^j$, and an accordion path of length $\ell - 3$ from w_1^i to y_k^j ; if a_i is negated in c_j , replace the edges $w_3^i s_k^j$ and $w_1^i x_k^j$ with $w_1^i s_k^j$ and $w_3^i x_k^j$.

The same argument of Propositions 10 and 11 shows that the resulting graph H has a C_ℓ -transverse matching if and only if the original instance of $\frac{(\ell+1)}{2}$ -SAT has a satisfiability function. When $\ell \geq 5$, $\frac{(\ell+1)}{2}$ -SAT is NP-complete. Therefore, the C_ℓ -transverse matching problem is NP-complete. $\square$

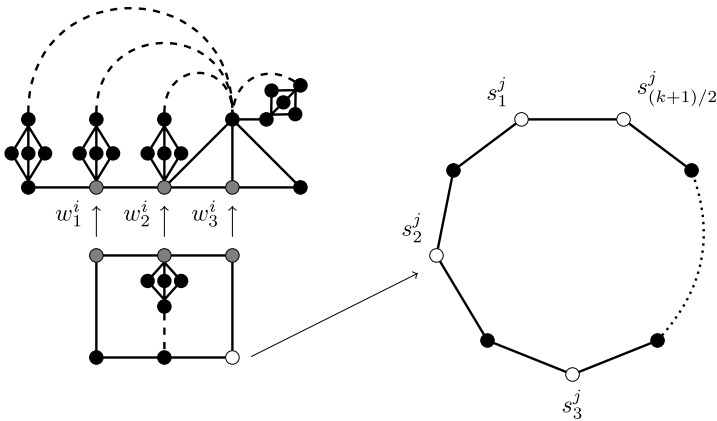


Fig. 4. Connecting the cycle corresponding to clause c_j to the variable gadget for a_i . Dashed lines in the left-hand gadgets indicate accordion paths.

5 Conclusion and Further Remarks

Theorem 1 follows from Corollary 6 and Propositions 10, 11, and 12. Theorem 2 follows from Theorem 1, Corollary 9, and the fact that $\text{ST}(C_\ell)$ for line graphs is equivalent to $\text{TM}(C_\ell)$ for each $\ell \geq 4$.

We have classified in this paper the complexity of the transverse matching problems for cycles of fixed lengths. Brandstädt [1] proved that determining whether a graph has a stable set which meets every cycle in the graph is an NP-complete problem even when it is restricted to bipartite graphs. We remark that an edge version of this problem is also NP-complete.

Consider the class $\mathcal{C}$ of subcubic planar graphs having exactly two vertices of degree two. It is an NP-complete problem to determine whether a graph in $\mathcal{C}$ contains a Hamiltonian path joining the two vertices of degree two [6]. Observe that such a path exists if and only if there is a matching that meets every cycle in the graph. Hence determining whether a graph in $\mathcal{C}$ contains a matching that meets every cycle in the graph is an NP-complete problem.

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